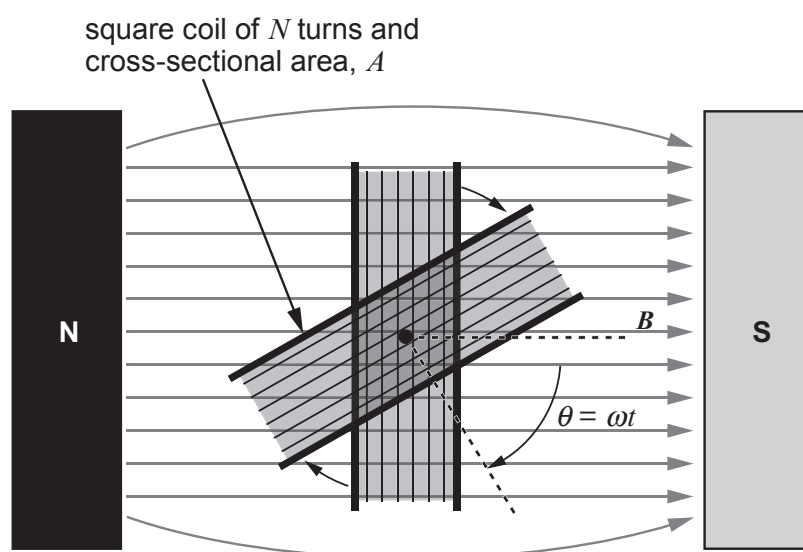


Option A – Alternating Currents

6. (a) A coil is rotated in a magnetic field.



- (i) Use Faraday's law to explain why the emf is proportional to the angular velocity of the coil. [2]

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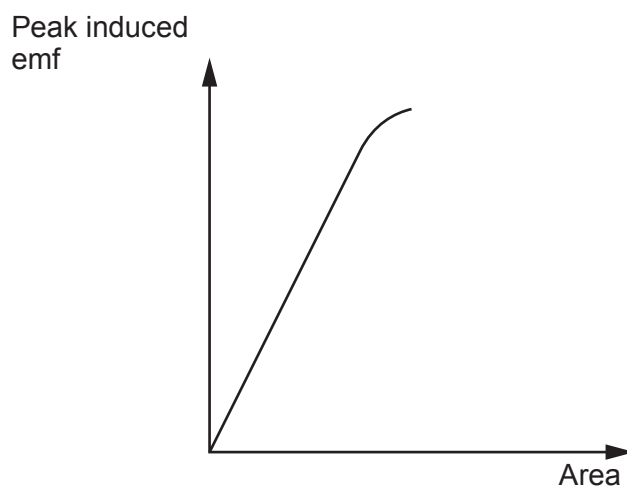
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The peak emf induced in a rotating coil is given by:

$$E = \omega BAN$$

An experiment is carried out to investigate the variation in the peak emf with area, A , by using coils with different cross-sectional areas. The following graph is obtained.



- (ii) Discuss the agreement of the graph with the equation for the peak emf and suggest a reason for any disagreement (see diagram of coil opposite). [3]

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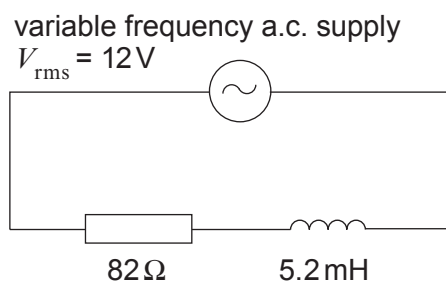
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- (b) (i) The minimum impedance of the circuit below occurs when the frequency of the a.c. supply is zero. Explain why this is so. [2]



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- (ii) Calculate the rms current when the rms pd across the resistor and inductor are equal. [3]

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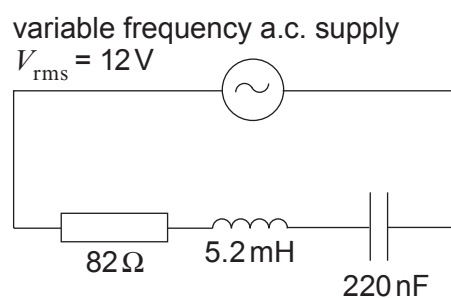
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- (c) (i) Explain how resonance arises in the LCR circuit shown. [3]



- (ii) Calculate the power dissipated in the circuit at resonance. [2]

- (iii) Calculate the rms current when the frequency of the supply is 9.40 kHz. [3]



- (iv) Discuss briefly the effectiveness of the given *LCR* circuit for providing a sharp resonance curve. [2]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$.)

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Examiner
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